

Activity- 3

AIM:

To draw the diagram of a given open circuit comprising at least a battery, resistor/rheostat, key, ammeter, and voltmeter. Mark the components that are not connected in the proper order and correct the circuit and the circuit diagram.

APPARATUS

A battery eliminator or battery (0 to 6V), rheostat, resistance box (0 to 100 ohms), two- or one-way key, D.C. ammeter (0-3A) and a DC voltmeter(0-3V).

THEORY:

An electrical circuit is functional only if all the components of the circuit are connected in proper order, assuming that all circuit components /devices are in working condition and the key is closed. An open circuit is the combination of primary components of an electric circuit in such a manner that on closing the circuit, no current is drawn from the battery.

PROCEDURE:

1. Draw a circuit diagram 1 and identify the various components that have not been connected in the proper order.
2. Draw the correct circuit diagram.
3. Ammeter: It should be connected in series with the battery eliminator
Voltmeter: It should be connected in parallel to the resistor
Rheostat: it should be connected in series (in place of the resistance coil) with the battery eliminator.
Resistance coil: It should be in parallel (in place of the rheostat)
One-way key: it should be connected in series of the battery eliminator
4. Connect the electrical components according to the corrected circuit diagram.
5. Close the key in the circuit to verify if the corrected circuit is functional.

RESULT:

The electrical circuit assembled as per the corrected circuit diagram is functional

Precautions

- i. The connections should be neat, clean, and tight.
- ii. All precautions regarding the polarity should be observed
- iii. Zero correction, if any, must be applied.

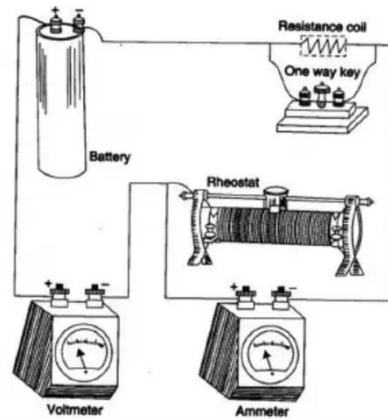
Sources of Error

- i. The instrument screws may be loose.
 - ii. Thick connecting wires may not be available.
 - iii. A rheostat may have high resistance.
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Diagram and observation table on the left side of the page

Diagram:

Open circuit diagram



Corrected circuit diagram

